

5.5.6

EE25BTECH11054 - S.Harsha Vardhan Reddy

Question:

For $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$, then $14\mathbf{A}^{-1}$ is given by

Solution:

Given

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} \quad (1)$$

Let $\mathbf{A}^{-1}$ be the inverse of $\mathbf{A}$. Then

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} \quad (2)$$

Augmented matrix of $(\mathbf{A} \mid \mathbf{I})$ is given by

$$\left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{array} \right) \quad (3)$$

Applying elementary row transformations:

$$\left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow 3R_2 + R_1} \left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 0 & 7 & 1 & 3 \end{array} \right) \quad (4)$$

$$\xrightarrow{R_1 \rightarrow 7R_1 - R_2} \left(\begin{array}{cc|cc} 21 & 0 & 6 & -3 \\ 0 & 7 & 1 & 3 \end{array} \right) \quad (5)$$

$$\xrightarrow{R_1 \rightarrow \frac{1}{21}R_1, R_2 \rightarrow \frac{1}{7}R_2} \left(\begin{array}{cc|cc} 1 & 0 & \frac{2}{7} & -\frac{1}{7} \\ 0 & 1 & \frac{1}{7} & \frac{3}{7} \end{array} \right) \quad (6)$$

Hence the inverse of matrix $\mathbf{A}$ is

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{2}{7} & -\frac{1}{7} \\ \frac{1}{7} & \frac{3}{7} \end{pmatrix} \quad (7)$$

Therefore, $14\mathbf{A}^{-1}$ is given by

$$14\mathbf{A}^{-1} = 14 \times \begin{pmatrix} \frac{2}{7} & -\frac{1}{7} \\ \frac{1}{7} & \frac{3}{7} \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ 2 & 6 \end{pmatrix} \quad (8)$$